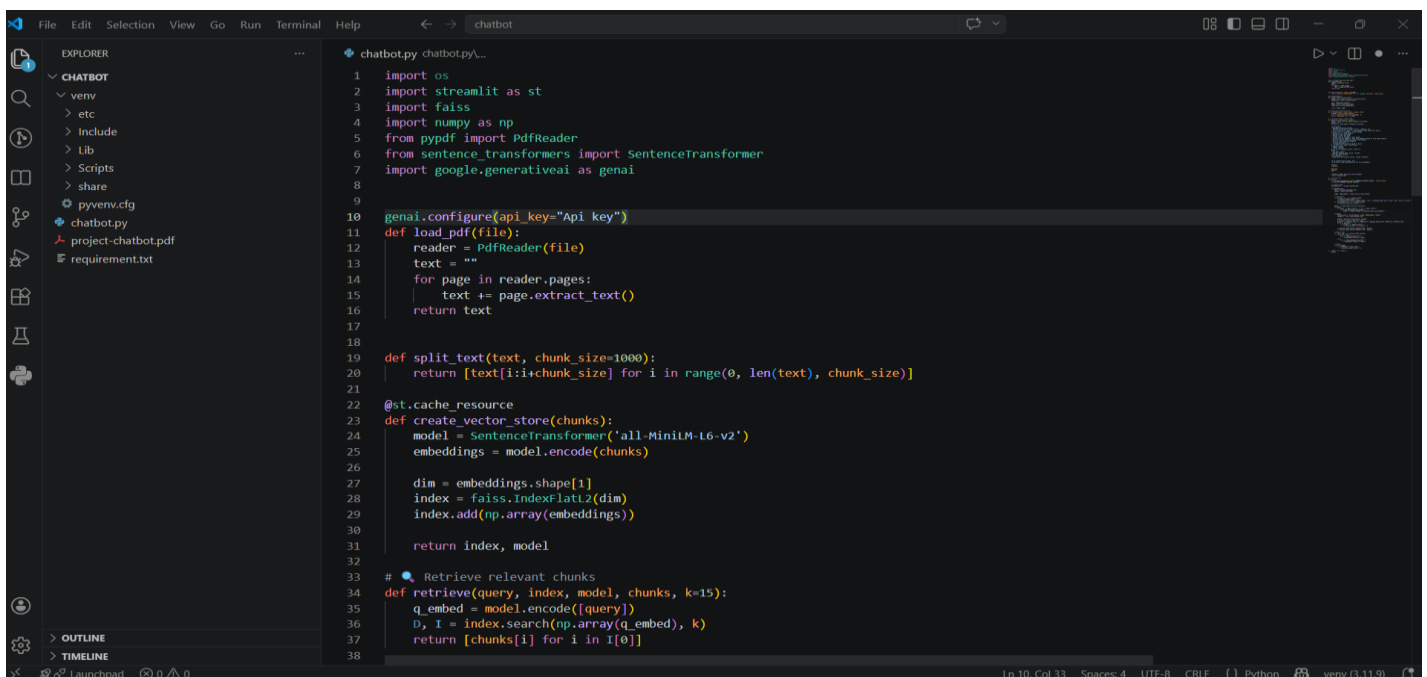


Myself Harshit Kushwaha ,I Have Successfully Made A Domain Specific Chatbot Using Rag , I Have Made An E-Commerce Advisor Chatbot Which Will Help Out The Customer To Find The Best Product Of Their Choice, My Chatbot Will Help The User Check The Price Of Product ,Comparision Between Different Product ,Payment Options For Product ,Ratings Of Items, Greeting The User At The Beginning Of Chat .I Have Used Google Gemini-2.5-Flash As Llm Model ,I Have given 200 product dummy data pdf which used for answering

CODE-



```
1 import os
2 import streamlit as st
3 import faiss
4 import numpy as np
5 from pypdf import PdfReader
6 from sentence_transformers import SentenceTransformer
7 import google.generativeai as genai
8
9
10 genai.configure(api_key="Api key")
11 def load_pdf(file):
12     reader = PdfReader(file)
13     text = ""
14     for page in reader.pages:
15         text += page.extract_text()
16     return text
17
18
19 def split_text(text, chunk_size=1000):
20     return [text[i:i+chunk_size] for i in range(0, len(text), chunk_size)]
21
22 @st.cache_resource
23 def create_vector_store(chunks):
24     model = SentenceTransformer('all-MiniLM-l6-v2')
25     embeddings = model.encode(chunks)
26
27     dim = embeddings.shape[1]
28     index = faiss.IndexFlatL2(dim)
29     index.add(np.array(embeddings))
30
31     return index, model
32
33 # Retrieve relevant chunks
34 def retrieve(query, index, model, chunks, k=15):
35     q_embed = model.encode([query])
36     D, I = index.search(np.array(q_embed), k)
37     return [chunks[i] for i in I[0]]
38
```

```
File Edit Selection View Go Run Terminal Help chatbot
EXPLORER
CHATBOT
  venv
  etc
  Include
  Lib
  Scripts
  share
  pyvenv.cfg
  chatbot.py
  project-chatbot.pdf
  requirement.txt
  OUTLINE
  TIMELINE
chatbot.py chatbot.py generate_answer
38
39 # Generate answer (STRICT RAG)
40 def generate_answer(query, context):
41     model = genai.GenerativeModel("gemini-2.5-flash")
42     prompt = f"""
43     You are an intelligent E-commerce assistant.
44
45     Instructions:
46     - Use ONLY the given context
47     - Understand the user's intent (price, category, etc.)
48     - If user asks for "under X", find products with price less than X
49     If user asks "is it good?", check ratings
50     - Rating > 4.0 → Good product
51     - Rating 3.0-4.0 → Average
52     - Rating < 3.0 → Not good
53     - Always mention ratings in your answer
54     - If user ask for "payment", then show payment option of just above product
55     - Extract product names and prices from context
56     - Give all best matching products
57     - If user ask-Tasks:
58       1. Recommend best products based on query
59       2. If user says "best", prioritize:
60         - Higher ratings
61         - Better features
62       3. If price condition exists, follow it
63     Rules:
64     - Use ONLY context
65     - Extract product name, price, ratings
66     - Show 2-3 best options
67     For comparison:
68     - Show clear differences (price, rating, features)
69
70
71     If no relevant data exists, say:
72     "I am sorry, that information is not in my database."
73
74     Context:
75     {context}
```

```
File Edit Selection View Go Run Terminal Help chatbot
EXPLORER
CHATBOT
  venv
  etc
  Include
  Lib
  Scripts
  share
  pyvenv.cfg
  chatbot.py
  project-chatbot.pdf
  requirement.txt
  OUTLINE
  TIMELINE
chatbot.py chatbot.py...
85 def main():
86     st.set_page_config(page_title="E-COMMERCE ADVISOR CHATBOT", layout="wide")
87     st.title("E-COMMERCE ADVISOR CHATBOT")
88
89     # Upload PDF
90     uploaded_file = "project-chatbot.pdf"
91
92     if uploaded_file:
93         text = load_pdf(uploaded_file)
94         chunks = split_text(text)
95
96         index, embed_model = create_vector_store(chunks)
97
98         # Chat memory
99         if "history" not in st.session_state:
100             st.session_state.history = []
101         st.session_state.history.append(("Bot", "Hi 🌟, Confused about what to buy? Don't worry I am here to help you, Please type the Question"))
102         if "last_product" not in st.session_state:
103             st.session_state.last_product = ""
104
105         query = st.chat_input("Ask your question")
106         if query:
107             if "above" in query.lower() or "it" in query.lower():
108                 if st.session_state.last_product:
109                     query = f"{query} about {st.session_state.last_product}"
110
111             if query:
112                 context_list = retrieve(query, index, embed_model, chunks)
113                 context = "\n\n".join(context_list)
114
115                 answer = generate_answer(query, context)
116                 # Detect product name from answer
117                 products = ["Nothing Phone 1", "iQOO Neo 7", "Samsung Galaxy S23", "iPhone 13", "OnePlus 11R"]
118                 for p in products:
119                     if p.lower() in answer.lower():
120                         st.session_state.last_product = p
121
122                 st.session_state.history.append(("You", query))
```

Output-

 Clear Chat

E-COMMERCE ADVISOR CHATBOT

 Bot: Hi 🤖, Confused about what to buy? Don't worry I am here to help you, Please type the Question below

 You: give me some phone under 25k

 Bot: Here are some phone options under ₹25,000:

1. **iPhone 13**
 - Price: Starting from ₹21,145 (on Flipkart)
 - Ratings: Flipkart 4.5/5 (Good), Myntra 4.6/5 (Good), Amazon 4.3/5 (Good)
 - Features: 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery
2. **OnePlus 11R**
 - Price: Starting from ₹24,270 (on Flipkart)
 - Ratings: Flipkart 4.8/5 (Good), Amazon 4.6/5 (Good), Myntra 4.1/5 (Good)
 - Features: 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery
3. **Samsung Galaxy S23**
 - Price: Starting from ₹11,446 (on Flipkart)
 - Ratings: Amazon 4.4/5 (Good), Flipkart 4.4/5 (Good), Myntra 4.1/5 (Good)
 - Features: 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery

 You: compare OnePlus 11R vs iphone13

 Bot: Here's a comparison between different options for OnePlus 11R and iPhone 13 based on the provided context:

OnePlus 11R Options:

1. **OnePlus 11R (Option 1):**
 - Price: Starting from ₹20362 (Flipkart)
 - Ratings: Up to 4.4/5 (Flipkart, Myntra) - A good product.
 - Features: OLED display, flagship processor, dual camera, fast charging.
2. **OnePlus 11R (Option 2):**
 - Price: Starting from ₹24270 (Flipkart)
 - Ratings: Excellent with up to 4.8/5 (Flipkart) - A good product.
 - Features: 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery.
3. **OnePlus 11R (Option 3):**
 - Price: Starting from ₹91616 (Flipkart)
 - Ratings: Excellent with up to 4.7/5 (Amazon) - A good product.
 - Features: 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery.

iPhone 13 Options:

1. **iPhone 13 (Option 1):**
 - Price: Starting from ₹21145 (Flipkart)
 - Ratings: Very good with up to 4.6/5 (Myntra) - A good product.
 - Features: 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery.
2. **iPhone 13 (Option 2):**
 - Price: Starting from ₹102757 (Flipkart)

- **Ratings:** Good with up to 4.5/5 (Flipkart) - A good product.
 - **Features:** 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery.
3. **iPhone 13 (Option 3):**
- **Price:** Starting from ₹116107 (Flipkart)
 - **Ratings:** Good with up to 4.4/5 (Amazon) - A good product.
 - **Features:** 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery.

Clear Differences & Comparison:

- **Price & Features:**
 - For similar budget-friendly options, the **OnePlus 11R (Option 1)** is the most affordable at ₹20362 with OLED display, flagship processor, dual camera, and fast charging (rated 4.4/5). Closely priced is **iPhone 13 (Option 1)** at ₹21145, offering 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery (rated 4.6/5). The iPhone 13 is slightly more expensive but has a higher rating in this specific comparison.
 - Moving to slightly higher prices, **OnePlus 11R (Option 2)** at ₹24270 has excellent ratings (4.8/5) and features 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery.
 - For a more premium segment, the **OnePlus 11R (Option 3)** is available from ₹91616 with a high rating of 4.7/5 and features 6.5-inch AMOLED, 5G, 8GB RAM, 128GB storage, 5000mAh battery. The **iPhone 13 (Option 2)** is priced higher at ₹102757 with a 4.5/5 rating and identical listed specifications.
 - The most expensive iPhone 13 in the context, **iPhone 13 (Option 3)**, is priced from ₹116107 with a 4.4/5 rating and the same listed specifications.
- **Ratings:** Both OnePlus 11R and iPhone 13 have multiple listings, all generally considered "good products" with ratings mostly above 4.0/5. The highest-rated individual listing for OnePlus 11R is 4.8/5, while for iPhone 13 it's 4.6/5.



You: suggest me some jeans



Bot: Here are some Levi's 511 Slim Jeans options available in our database:

1. **Levi's 511 Slim Jeans (Product ID 143)**
 - **Lowest Price:** ₹15182 on Flipkart
 - **Ratings:** Amazon 4.3/5 (Good), Flipkart 3.8/5 (Average), Myntra 3.8/5 (Average)
 - **Specifications:** Lightweight, comfortable, premium material
2. **Levi's 511 Slim Jeans (Product ID 111)**
 - **Lowest Price:** ₹32568 on Flipkart
 - **Ratings:** Amazon 3.9/5 (Average), Flipkart 4.6/5 (Good), Myntra 3.8/5 (Average)
 - **Specifications:** Cotton fabric, slim fit, breathable, durable stitching
3. **Levi's 511 Slim Jeans (Product ID 141)**
 - **Lowest Price:** ₹91531 on Flipkart
 - **Ratings:** Amazon 4.0/5 (Good), Flipkart 4.1/5 (Good), Myntra 3.8/5 (Average)
 - **Specifications:** Cotton fabric, slim fit, breathable, durable stitching
4. **Levi's 511 Slim Jeans (Product ID 136)**
 - **Lowest Price:** ₹103626 on Flipkart
 - **Ratings:** Amazon 4.2/5 (Good), Flipkart 4.8/5 (Good), Myntra 4.6/5 (Good)
 - **Specifications:** Cotton fabric, slim fit, breathable, durable stitching



You: give some high specifications laptop under 50k



Bot: Here are some high-specification laptops available for under ₹50,000, based on your criteria:

1. **Lenovo ThinkPad E14**
 - **Price:** ₹42,474 (on Flipkart)

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- **Specifications:** Ryzen 7 processor, 16GB RAM, SSD storage, lightweight design
 - **Ratings:** Amazon 4.7/5, Flipkart 4.7/5, Myntra 4.6/5 (All ratings are good, making it an excellent product.)
2. **HP Victus 16**
- **Price:** ₹25,557 (on Flipkart)
 - **Specifications:** Intel i5/i7 processor, 16GB RAM, 512GB SSD, Windows 11
 - **Ratings:** Amazon 4.7/5, Flipkart 4.4/5, Myntra 4.2/5 (All ratings are good, making it an excellent product at a very competitive price.)
3. **Asus ROG Zephyrus G14**
- **Price:** ₹41,893 (on Flipkart)
 - **Specifications:** Ryzen 7 processor, 16GB RAM, SSD storage, lightweight design
 - **Ratings:** Amazon 3.9/5, Flipkart 4.4/5, Myntra 4.7/5 (Ratings are average to good, with excellent ratings on Flipkart and Myntra.)



You: should i buy HP Victus 16 or Asus ROG Zephyrus G14



Bot: Here are the best matching products from HP Victus 16 and Asus ROG Zephyrus G14 for your consideration, based on higher ratings and features:

1. **HP Victus 16 (Ryzen 7, ID 77)**
- **Price:** Starting from **₹9347** on Flipkart.
 - **Ratings:** Amazon 4.9/5 (Good), Flipkart 4.0/5 (Average), Myntra 4.5/5 (Good). **Overall: Good product.**
 - **Features:** Ryzen 7 processor, 16GB RAM, SSD, lightweight design.
 - **Key Difference:** This HP Victus 16 model offers exceptional ratings at a remarkably low price, making it a great value for a powerful Ryzen 7 system.
2. **HP Victus 16 (Intel i5/i7, ID 59)**
- **Price:** Starting from **₹25557** on Flipkart.
 - **Ratings:** Amazon 4.7/5 (Good), Flipkart 4.4/5 (Good), Myntra 4.2/5 (Good). **Overall: Good product.**
 - **Features:** Intel i5/i7 processor, 16GB RAM, 512GB SSD, Windows 11.
 - **Key Difference:** This model provides a robust Intel-based configuration with very good ratings, suitable if you prefer an Intel CPU and dedicated SSD storage, at a competitive mid-range price.
3. **Asus ROG Zephyrus G14 (Intel i5/i7, ID 63)**
- **Price:** Starting from **₹38181** on Flipkart.
 - **Ratings:** Amazon 4.7/5 (Good), Flipkart 4.6/5 (Good), Myntra 3.7/5 (Average). **Overall: Good product.**
 - **Features:** Intel i5/i7 processor, 16GB RAM, 512GB SSD, Windows 11.
 - **Key Difference:** This Asus ROG Zephyrus G14 offers similar core specifications (Intel i5/i7, 16GB RAM, 512GB SSD) to the HP Victus 16 (ID 59) but comes at a higher price point. It has excellent ratings on Amazon and Flipkart.

Comparison Summary:

- If you are looking for the most **budget-friendly option with great performance and high ratings**, the **HP Victus 16 (ID 77)** is an outstanding choice with a Ryzen 7 processor for just **₹9347**.
- If you prefer an **Intel processor** and a balanced mid-range price, the **HP Victus 16 (ID 59)** at **₹25557** offers excellent ratings and a 512GB SSD.
- The **Asus ROG Zephyrus G14 (ID 63)** at **₹38181** is another strong Intel-based contender with a 512GB SSD and good ratings, though at a higher price than the similarly configured HP Victus 16 (ID 59).



You: what are the payment option for Asus ROG Zephyrus G14 (Intel i5/i7, ID63)



Bot: The payment options for Asus ROG Zephyrus G14 (Intel i5/i7) are:

- UPI
- Credit/Debit Cards

- EMI
- COD (Cash on Delivery)
- Net Banking

This product has ratings of Amazon 4.7/5, Flipkart 4.6/5, and Myntra 3.7/5, making it a good product overall.



You: suggest some washing machine



Bot: Here are some washing machine recommendations based on your request:

1. Samsung 6.5kg Washing Machine

- **Prices:** Available from ₹54931 (Flipkart) to ₹57338 (Myntra).
- **Ratings:** Amazon 4.2/5, Flipkart 4.6/5, Myntra 4.5/5. This is a **good product**.
- **Features:** High capacity, digital control, modern design.

2. Panasonic Washing Machine

- **Prices:** Available from ₹40059 (Flipkart) to ₹43011 (Myntra).
- **Ratings:** Amazon 4.3/5, Flipkart 4.4/5, Myntra 4.1/5. This is a **good product**.
- **Features:** High capacity, digital control, modern design.

3. Samsung 6.5kg Washing Machine

- **Prices:** Available from ₹26612 (Flipkart) to ₹28705 (Myntra).
- **Ratings:** Amazon 4.2/5, Flipkart 3.8/5, Myntra 4.3/5. This is a **good product**.
- **Features:** High capacity, digital control, modern design.

Comparison: The first Samsung 6.5kg Washing Machine (priced around ₹55k-₹57k) has the highest overall ratings among these options (up to 4.6/5). The Panasonic Washing Machine (around ₹40k-₹43k) also has excellent ratings (up to 4.4/5) and similar features. If you are looking for a more budget-friendly option, the second Samsung 6.5kg Washing Machine (around ₹26k-₹28k) is a good choice with solid ratings (up to 4.3/5) and the same core features.

Ask your question

